Lab 3: GO NETWORK PROGRAMMING

Net-Centric Programming

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Part 1: TCP Basic – Echo Server

Task 1.1: TCP Chat room server

```
thanhphatchau@Ants-MacBook-Pro lab3 % go run main.go ser
ver
Chat server listening on :9000
Alice joined the chat
Bob joined the chat
Broadcasting from Alice: Hello Everyone
Broadcasting from Bob: Hi Alice!
Broadcasting from Alice: Hi Bob! how are you
Broadcasting from Bob: good
Bob left the chat
```

```
thanhphatchau@Ants-MacBook-Pro lab3 % go run main.go cli
ent Alice
Enter username: Alice
Type messages (type 'exit' to quit):
*** Alice joined the chat ***
*** Bob joined the chat ***
Hello Everyone
[Bob]: Hi Alice!
Hi Bob! how are you
[Bob]: good
exit*** Bob left the chat ***
```

```
thanhphatchau@Ants-MacBook-Pro lab3 % go run main.go cli
ent Bob
Enter username: Bob
Type messages (type 'exit' to quit):
*** Bob joined the chat ***
[Alice]: Hello Everyone
Hi Alice!
[Alice]: Hi Bob! how are you
good
exit
Leaving chat...
thanhphatchau@Ants-MacBook-Pro lab3 % 
```

Task 1.2: TCP File transfer

```
thanhphatchau@Ants-MacBook-Pro task12 % go run main.go c
lient document.txt

Sending file: document.txt
Connected to server

✓ File transferred successfully!
thanhphatchau@Ants-MacBook-Pro task12 % 
```

```
thanhphatchau@Ants-MacBook-Pro task12 % go run main.go s
erver
File Transfer Server listening on :8081
Client connected
Receiving file: document.txt (0 bytes)
File saved successfully to received/document.txt
```

Part 2: UDP Basics – Message Broadcasting

Task 2.1: UDP Ping Monitor

```
thanhphatchau@Ants-MacBook-Pro task21 % go run main.go
=== UDP Ping Monitor ===
Starting ping servers on ports: 9001, 9002, 9003
server-2 listening on :9002
server-1 listening on :9001
Pinging servers...
Server               Status     RTT
------------------------------------------
localhost:9001       ✓ Online   130.917µs
localhost:9002       ✓ Online   377.417µs
localhost:9003       ✗ Timeout  -
Average RTT: 254.167µs
Success Rate: 66.67% (2/3)

Pinging servers...
Server               Status     RTT
------------------------------------------
localhost:9001       ✓ Online   150.792µs
localhost:9002       ✓ Online   160.75µs
localhost:9003       ✗ Timeout  -
Average RTT: 155.771µs
Success Rate: 66.67% (2/3)

Pinging servers...
Server               Status     RTT
------------------------------------------
localhost:9001       ✓ Online   146.375µs
localhost:9002       ✓ Online   120µs
localhost:9003       ✗ Timeout  -
Average RTT: 133.187µs
Success Rate: 66.67% (2/3)

Pinging servers...
Server               Status     RTT
------------------------------------------
localhost:9001       ✓ Online   125.833µs
localhost:9002       ✓ Online   169.542µs
localhost:9003       ✗ Timeout  -
Average RTT: 147.687µs
Success Rate: 66.67% (2/3)
```

Task 2.2: UDP Discovery Service

```
thanhphatchau@Ants-MacBook-Pro task22 % go run main.go
=== Service Discovery ===
Starting services:
- Database Service on port 5432
Discovery listener running on :8083
- Web Service on port 8080
- API Service on port 3000
Discovering services...
Found 3 services:
Service Name          Address          Port
------------------------------------------------
API Service           192.168.101.29   3000
Database Service      192.168.101.29   5432
Web Service           192.168.101.29   8080
Discovery complete!
```

Part 3: TCP vs UDP Comparison

Task 3.1: Performance Comparison tool

```
thanhphatchau@Ants-MacBook-Pro task31 % go run main.go
UDP test server listening on :9101
TCP test server listening on :9100
=== TCP vs UDP Performance Comparison ===
Testing data transfer speeds...
Data Size   | TCP Avg Time | UDP Avg Time | UDP Faster By
------------|--------------|--------------|---------------
1 KB        | 515.00 µs    | 180.00 µs    | 65%
10 KB       | 281.00 µs    | 423.00 µs    | -51%
100 KB      | 404.00 µs    | 2.19 ms      | -441%
1 MB        | 812.00 µs    | 9.78 ms      | -1104%
Summary:
- UDP is usually faster than TCP in this local benchmark
- Average speed improvement: -382.8%
- TCP provides reliability, UDP provides speed
```

Part 4: RWA – Multi Protocol Chat

Task 4.1: Hybrid chat server

```
thanhphatchau@Ants-MacBook-Pro task41 % go run main.go c
lient Bob
Status Update: Bob is online
[09:04:28] Bob: /say hi
/away
Status Update: Bob is away
Status Update: Bob is online
/history 65
[09:03:40] Alice: hello everyonehello
[09:03:52] Alice: /history
[09:04:16] Bob: hello Alice
[09:04:23] Bob: What are u doing
[09:04:28] Bob: /say hi
Status Update: Bob is online
Status Update: Bob is online
exit
Leaving chat...
```

```
thanhphatchau@Ants-MacBook-Pro task41 % go run main.go c
lient Alice
Status Update: Alice is typing...
Status Update: Alice is online
[09:03:52] Alice: /history
Status Update: Alice is online
Status Update: Bob is online
Status Update: Alice is online
Status Update: Bob is online
/aStatus Update: Alice is online
wayStatus Update: Bob is online

Status Update: Alice is away
Status Update: Alice is online
Status Update: Bob is online
exit
Leaving chat...
thanhphatchau@Ants-MacBook-Pro task41 %
```

```
thanhphatchau@Ants-MacBook-Pro task41 % go run main.go s
erver
[UDP] Bob: online
[UDP] Alice: online
[UDP] Bob: online
[UDP] Alice: away
[UDP] Alice: online
[UDP] Bob: online
[UDP] Alice: away
Alice disconnected
[UDP] Bob: online
[TCP] Bob: hello Alice
[UDP] Bob: typing
[UDP] Bob: online
[UDP] Bob: online
[UDP] Bob: typing
[UDP] Bob: online
[TCP] Bob: What are u doing
```

Part 5: Network Protocol Design

Task 5.1: Custom protocol implementation

```
thanhphatchau@Ants-MacBook-Pro task51 % go run main.go
UDP test server listening on :9101
TCP test server listening on :9100
=== TCP vs UDP Performance Comparison ===
Testing data transfer speeds...
Data Size   | TCP Avg Time  | UDP Avg Time  | UDP Faster By
------------|---------------|---------------|---------------
1 KB        | 475.00 µs     | 164.00 µs     | 65%
10 KB       | 313.00 µs     | 362.00 µs     | -15%
100 KB      | 415.00 µs     | 2.29 ms       | -452%
1 MB        | 635.00 µs     | 8.91 ms       | -1304%
Summary:
- UDP is usually faster than TCP in this local benchmark
- Average speed improvement: -426.5%
- TCP provides reliability, UDP provides speed
```